

MENGSHAN, China

WMO: 590580

Lat: 24.2036N

Lon: 110.5109E

Elev: 183

StdP: 99.14

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.5	3.8	-4.6	2.6	9.7	-2.2	3.2	8.4	6.2	7.1	5.5	8.4	2.3	320	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	34.6	25.7	33.7	25.6	32.8	25.6	27.2	31.8	26.9	31.2	26.5	30.6	2.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	22.0	29.4	25.7	21.5	28.9	25.4	21.1	28.6	87.9	32.0	86.0	31.3	84.3	30.9	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.5	4.1	0.5	36.8	1.1	0.9	-0.3	37.5	-0.9	38.0	-1.6	38.6	-2.4	39.3		
			-0.3	27.8	1.3	0.7	-1.2	28.2	-2.0	28.6	-2.7	29.0	-3.6	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	10.4	12.6	15.8	21.1	25.0	27.1	28.4	26.4	22.9	17.8	12.5	(6)	(7)
		DBStd	7.14	4.17	5.01	4.19	3.72	2.49	1.82	1.56	1.65	2.36	2.91	3.48		
		HDD10.0	89	46	26	4	0	0	0	0	0	0	1	12		
		HDD18.3	772	246	172	100	17	0	0	0	0	4	50	183		
		CDD10.0	4011	59	98	185	332	464	514	570	493	399	234	91		
		CDD18.3	1652	1	10	23	99	206	264	312	312	243	145	3		
		CDH23.3	14661	3	39	78	531	1546	2476	3354	3315	2188	951	173		
(14)	Wind	CDH26.7	5211	0	4	8	116	456	814	1340	1353	819	277	23	(14)	(15)
		WSAvg	1.7	2.1	2.0	1.8	1.6	1.5	1.4	1.4	1.4	1.6	1.8	1.9		
		PrecAvg	1691	50	78	96	208	293	270	200	205	101	87	62		
		PrecMax	2529	143	235	219	435	608	532	581	419	214	220	173		
		PrecMin	1147	2	5	27	66	93	36	61	5	10	5	0		
		PrecStd	312	39	59	48	80	112	131	133	101	50	67	52		
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	27.1	27.5	31.3	33.4	34.2	36.2	36.0	34.7	32.7	29.1	(20)	(21)
			MCWB	17.8	20.2	21.6	24.0	24.9	25.9	25.8	25.7	25.2	23.3	21.3		
		2%	DB	21.0	24.4	25.3	29.6	32.0	33.0	34.5	34.6	33.4	31.0	27.1		
			MCWB	16.0	18.8	20.3	23.2	24.6	26.0	25.9	25.7	25.0	22.4	19.5		
		5%	DB	19.0	22.0	23.6	28.0	30.7	32.0	33.4	33.6	32.4	29.5	25.4		
			MCWB	14.8	18.0	19.5	22.8	24.3	25.8	26.0	25.7	24.8	22.0	18.9		
		10%	DB	17.0	20.0	21.8	26.4	29.3	30.8	32.3	32.5	31.2	28.2	23.6		
			MCWB	13.8	17.0	18.8	22.3	24.2	25.7	25.9	25.8	24.5	21.3	18.1		
		0.4%	WB	19.0	21.1	22.9	25.6	27.0	27.6	27.8	27.7	27.0	25.3	23.1		
			MCDB	22.0	26.0	26.3	29.2	30.7	31.8	32.8	32.5	31.8	29.9	26.6		
		2%	WB	17.4	19.7	21.8	24.6	26.2	27.0	27.2	27.2	26.4	24.6	22.0		
			MCDB	20.0	22.9	24.4	27.9	29.8	31.0	31.8	31.9	30.9	28.5	24.9		
(35)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	15.9	18.6	20.6	23.8	25.6	26.6	26.7	26.7	25.9	23.8	20.9	(35)	(36)
			MCDB	18.1	21.2	22.5	26.7	29.0	30.3	31.1	31.2	30.1	27.2	23.8		
		10%	WB	14.0	17.3	19.4	23.1	25.0	26.1	26.3	26.3	25.4	22.9	19.5		
			MCDB	16.3	19.9	21.3	25.5	28.2	29.6	30.5	30.5	29.2	25.9	22.2		
		5% DB	MDBR	6.1	6.0	5.7	6.0	6.7	6.1	7.0	7.4	7.8	8.3	8.2		
			MCDBR	11.3	10.0	9.4	8.9	9.2	8.3	9.2	9.5	9.8	10.6	11.5		
			MCWBR	6.1	5.0	4.7	3.8	3.4	2.7	2.4	2.5	2.9	3.6	4.6		
			MCDBR	8.5	7.8	6.7	6.8	6.8	6.7	7.8	8.2	8.1	7.9	7.7		
		5% WB	MCWBR	5.3	4.6	4.3	3.9	3.1	2.7	2.6	2.7	2.8	3.3	4.0		
		taub		0.645	0.692	0.830	0.795	0.633	0.594	0.565	0.637	0.679	0.724	0.613		
				1.602	1.563	1.370	1.467	1.801	1.911	1.984	1.778	1.669	1.539	1.713		
				610	621	564	603	708	731	752	699	655	589	632		
				238	264	334	309	220	196	182	223	244	266	211		
(44)	All-Sky Solar Radiation	RadAvg	2.03	2.18	2.40	3.03	3.76	3.92	4.78	4.69	4.30	3.83	2.95	2.46	(44)	(45)
		RadStd	0.51	0.47	0.41	0.43	0.39	0.56	0.47	0.32	0.35	0.36	0.51	0.55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	+0.48	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+0.42	N/A	N/A	N/A	+88	+65

Nomenclature: See separate page